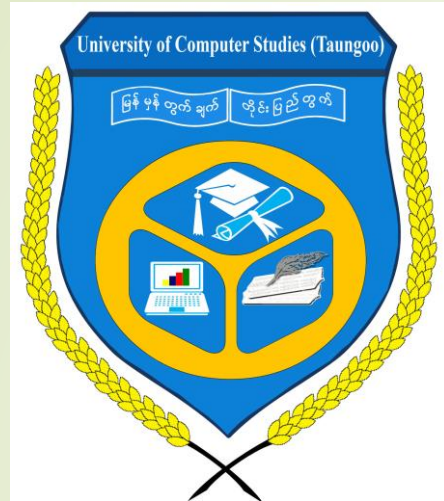




Introduction To Python

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Contents

- What is Python?
- History of Python
- Installation steps for Python Environment
- Basic Elements of Python
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- Summary

Loops and Decision Making

Decision Making

- If
- If ... else
- If elif...else
- Nested if

Loops

- for
- while
- Nested for

Loops and Decision Making

➤ Python supports the usual logic conditions from mathematics:

Equal	: <code>a==b</code>
Not Equal	: <code>a!=b</code>
Less than	: <code>a<b</code>
Less than or equal to	: <code>a<=b</code>
Greater than	: <code>a>b</code>
Greater than or equal to	: <code>a>=b</code>

Decision Making

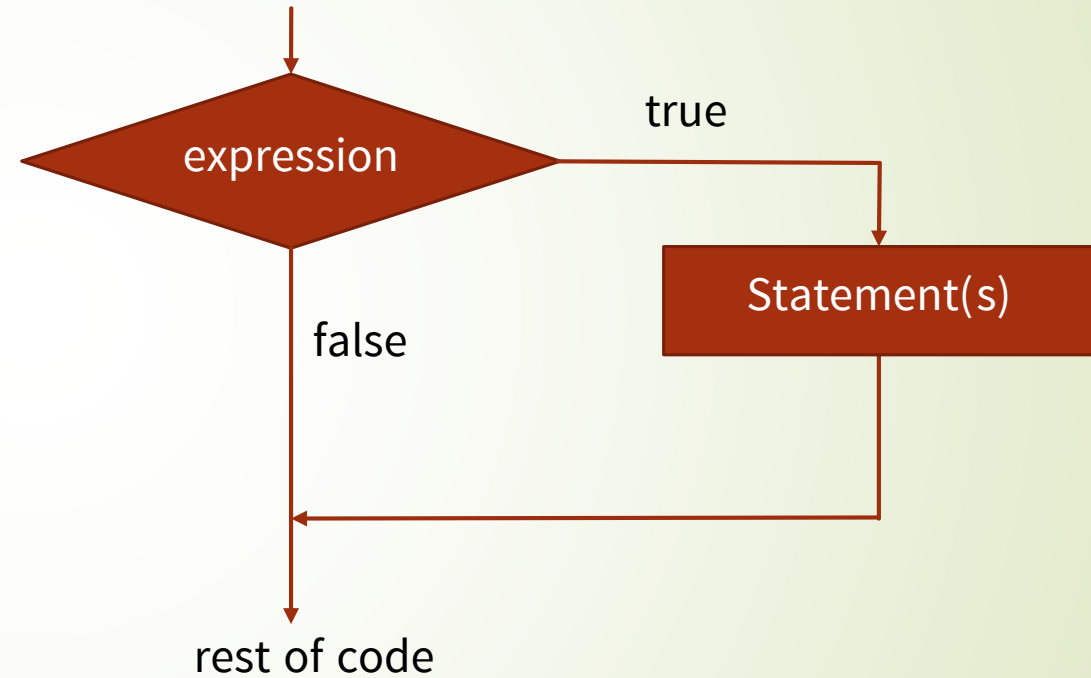
If statement

➡ Syntax

if expression :
 statement(s)

Or

if (expression):
 statement(s)



Decision Making

if statement

Example:

```
n= int(input("Please enter a number = "))
```

```
if (n>0):
```

```
    print (n, " is positive number")
```

```
if (n<0):
```

```
    print(n, " is negative number")
```

Output:

```
Please enter a number = 5  
5 is positive number
```

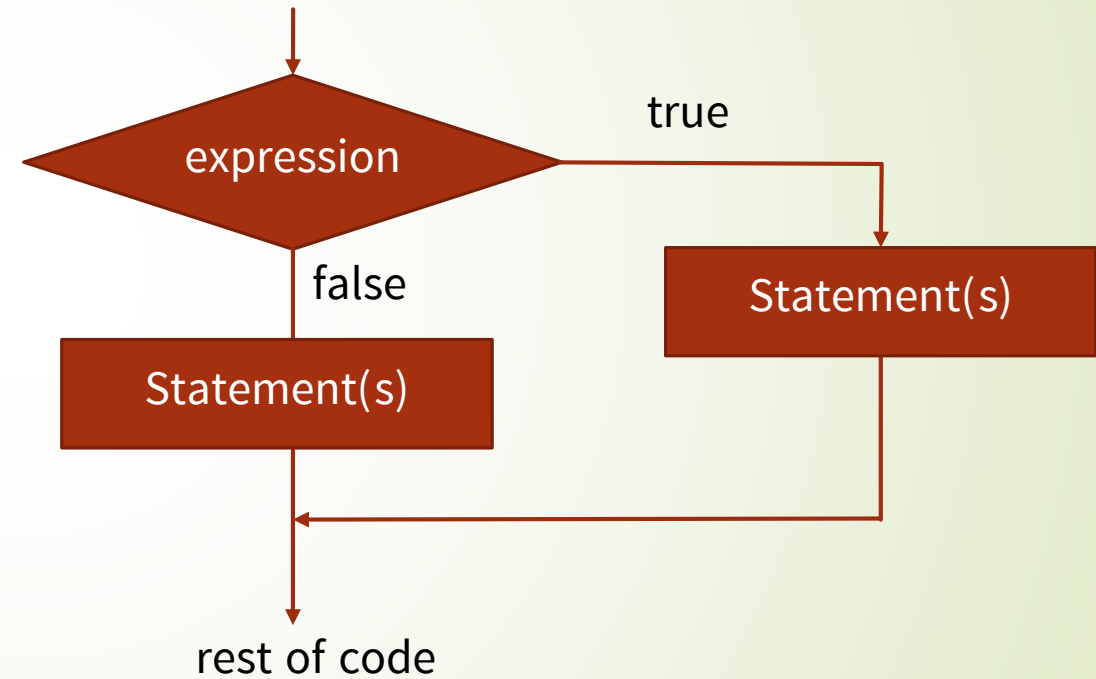
Decision Making

Ifelse statement

➤ **Syntax**

```
if expression:  
    statement(s)
```

```
else:  
    statement(s)
```



Decision Making

if else statement

Example:

```
n= int(input("Please enter a number = "))
```

```
r = n%2
```

```
if (r==0):
```

```
    print (n, " is event number")
```

```
else:
```

```
    print(n, " is odd number")
```

Output:

Please enter a number = 5

5 is odd number

Decision Making

ifelif...else statement

➤ **Syntax**

if expression :

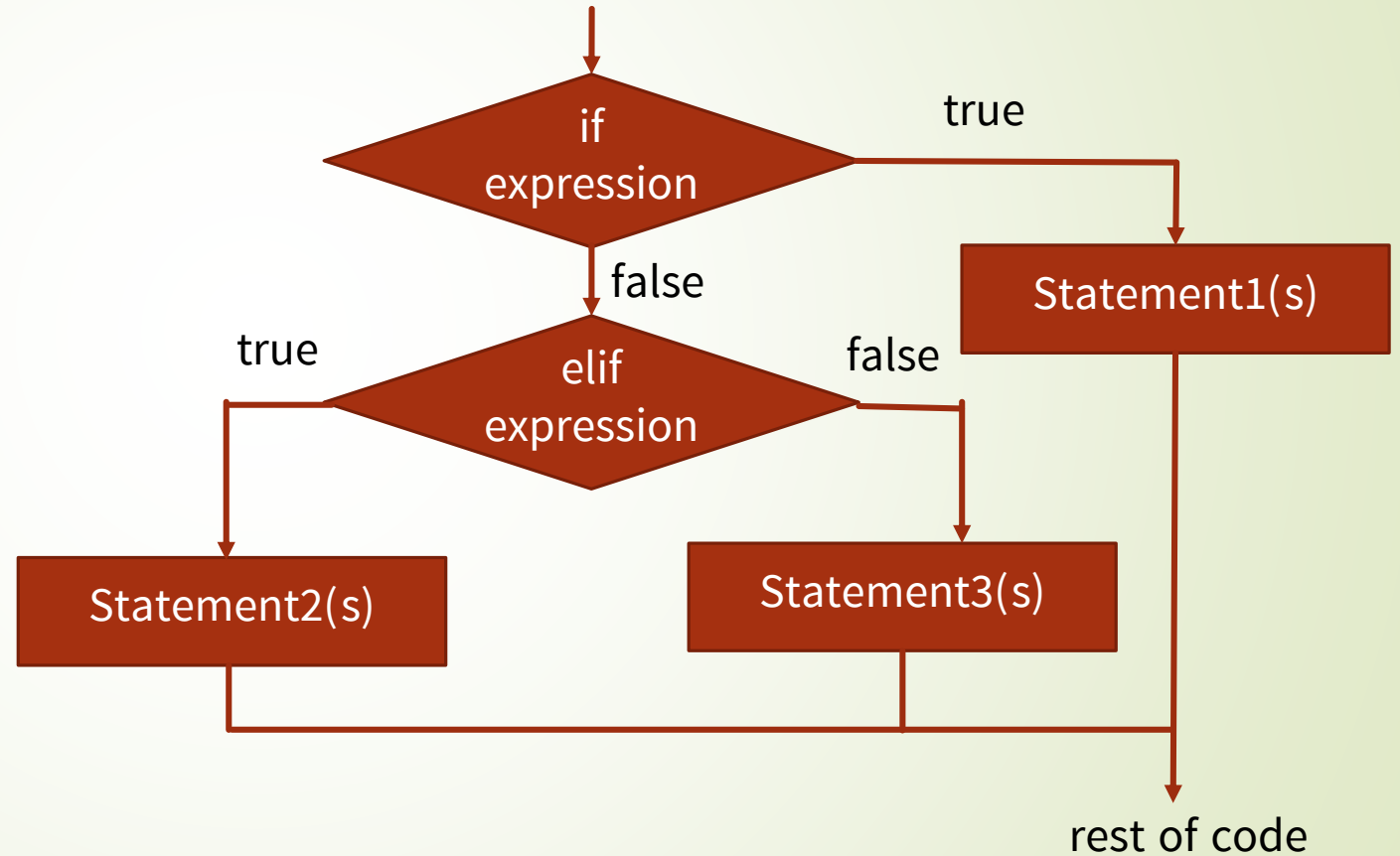
statement1(s)

elif (expression):

statement2(s)

else :

statement3(s)



Decision Making

if elif else statement

Example:

```
n= int(input("Please enter a number = "))
```

```
if (n==1):
```

```
    print (n, " is Monday")
```

```
elif (n==2):
```

```
    print(n, " is Tuesday")
```

```
elif (n==3):
```

```
    print(n, " is Wednesday")
```

```
else:
```

```
    print(n, " is one day of Thursday to Sunday")
```

Output:

Please enter a number = 5

5 is one day of Thursday to Sunday

Decision Making

if elif else statement

Q. Calculate the grade for a given mark.

Marks obtained	Grade
≥ 90	A
85-89	A-
80-84	B+
75-79	B
70-74	B-
65-69	C+
60-64	C
55-59	C-
50-54	D
0-49	F

Output:

Enter mark: 85

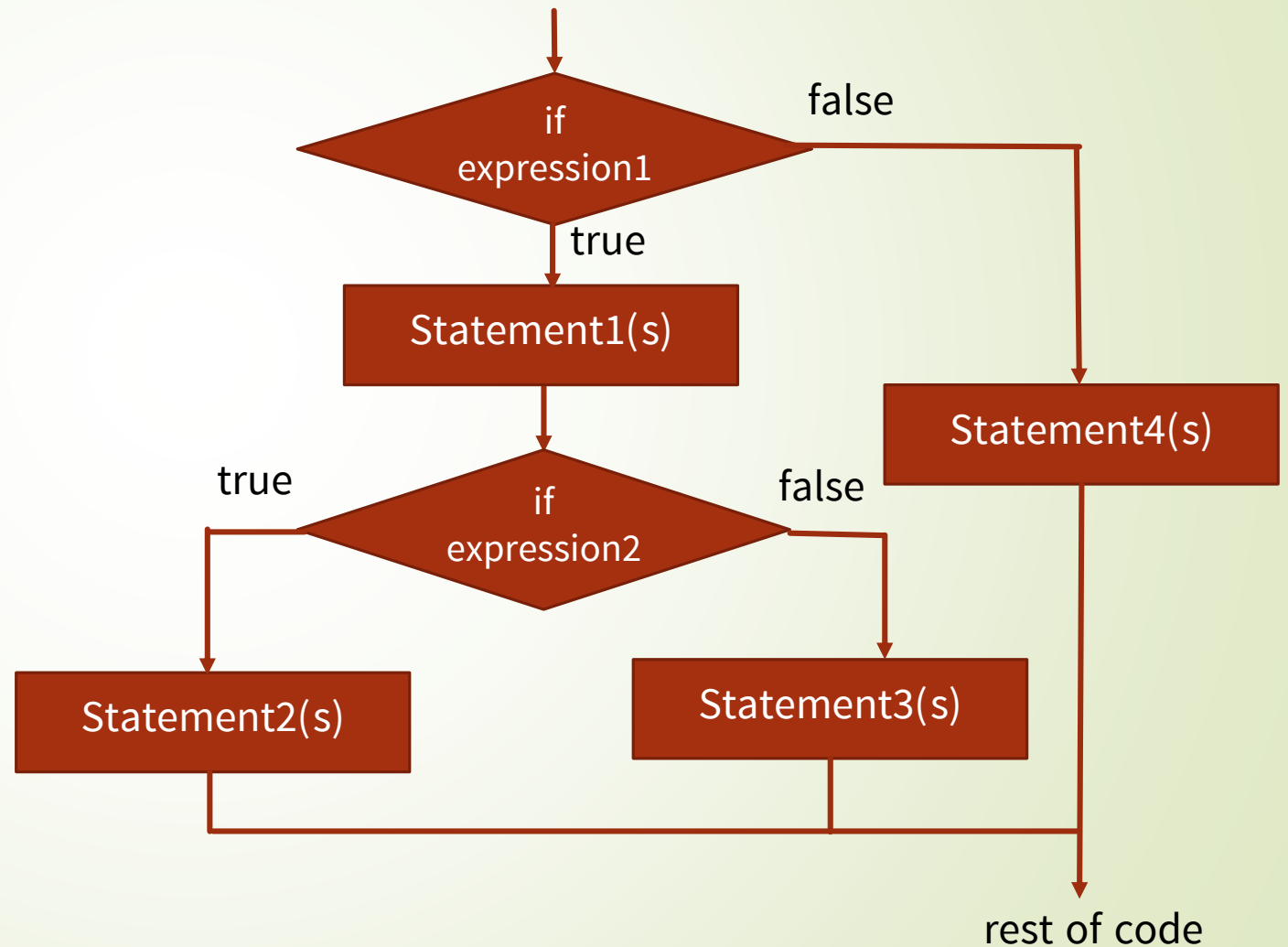
A-

Decision Making

Nested if statement

➤ Syntax

```
if expression1 :  
    statement1(s)  
    if (expression2):  
        statement2(s)  
    else :  
        statement3(s)  
else:  
    statement4(s)
```



Decision Making

Nested if statement

Example:

```
n= int(input("Please enter a number = "))
```

```
if (n>=0):
```

```
    if (n==0):
```

```
        print( "The number is Zero.")
```

```
    else:
```

```
        print( "The number is Positive.")
```

```
else:
```

```
    print( " The number is Negative.")
```

Output:

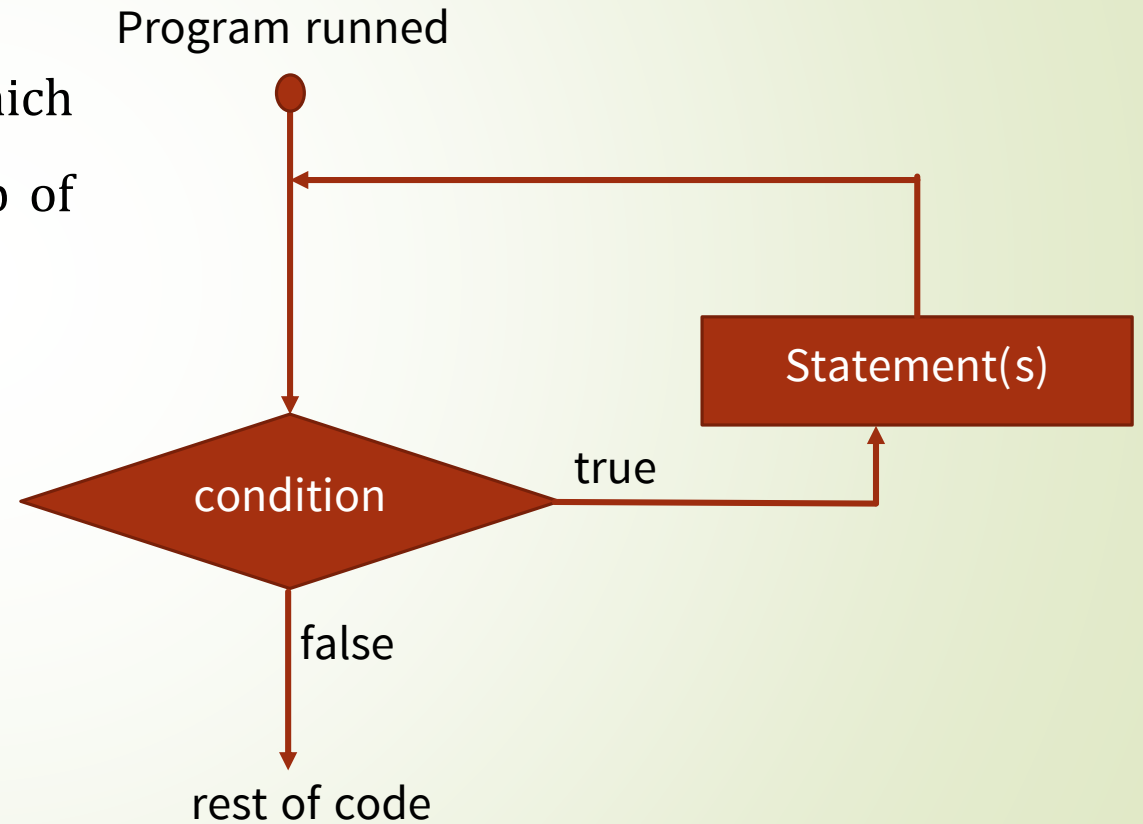
```
Please enter a number = 5  
The number is Positive
```

Loops

→ Loops are the the set of instruction which allows us to execute a statement or group of statements multiple times.

1) for loop

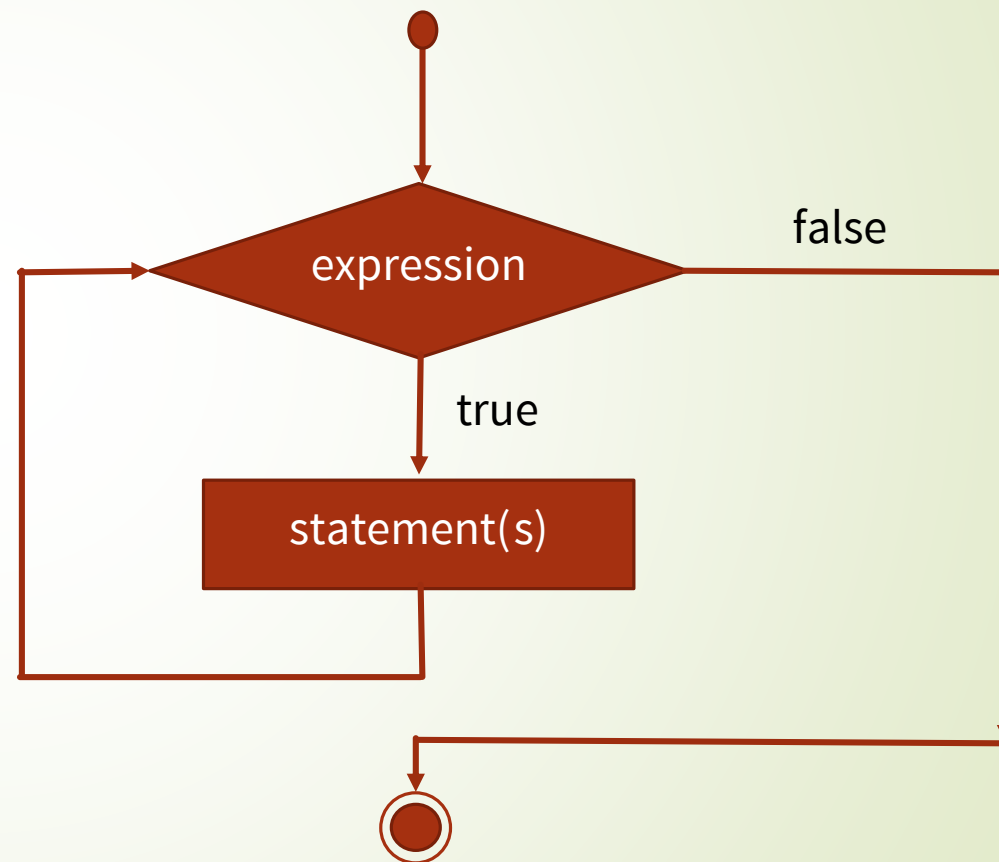
2) while loop



Loops

Flow Chart of “while loop”

while expression:
statement(s)



Loops

Example of “while loop”

Q. Write a program for sum of integer upto 'n' by using while loop.

Example:

```
n=int(input("Enter a number: "))
```

```
s=0
```

```
i=1
```

```
while (i<=n):
```

```
    s=s+i
```

```
    i=i+1
```

```
    print("Sum= ",s)
```

Output:

Enter a number = 5

Sum=1

Sum=3

Sum=6

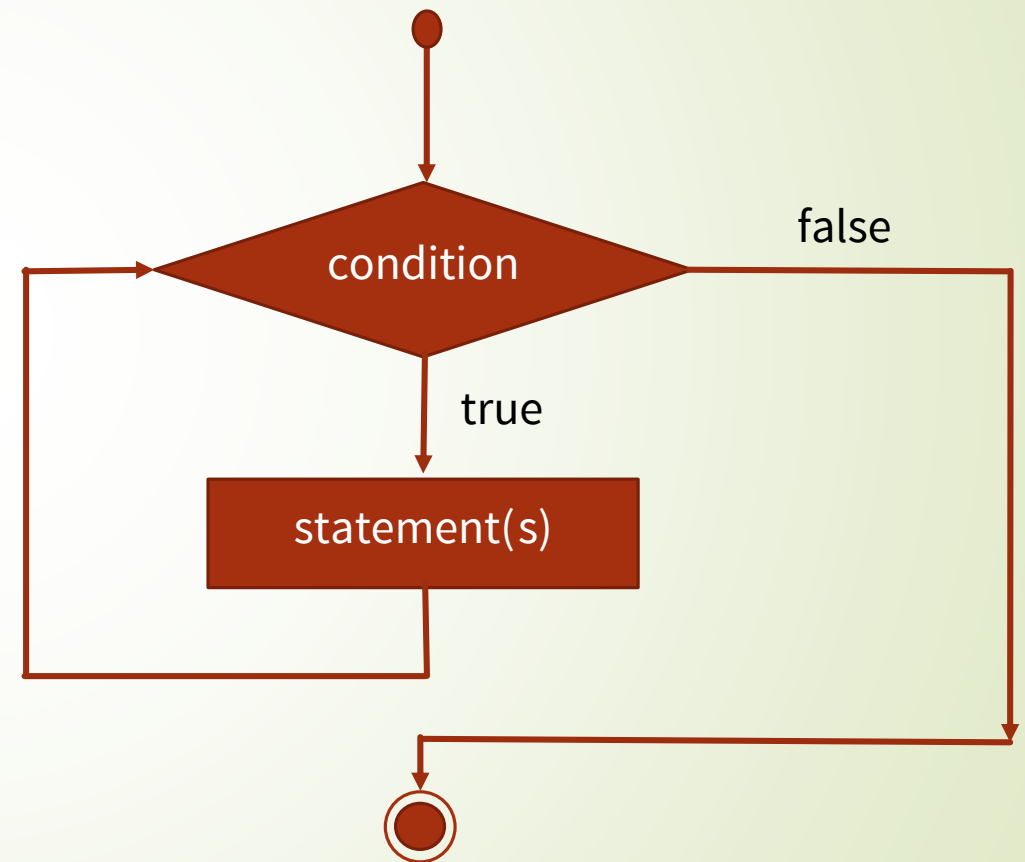
Sum=10

Sum=15

Loops

Flow Chart of “for loop”

for **var** **in** **sequence**:
statement(s)



Loops

Example of “for loop”

Q. Write a program for sum of integer upto 'n' by using for loop.

Example1:

```
n=int(input("Enter a number: "))  
s=0  
for i in range(1, n+1, 1):  
    s=s+i  
    print("sum= ", s)
```

Example2:

```
for x in "banana":  
    print(x)
```

Output1:

```
Enter a number = 5  
sum=1  
sum=3  
sum=6  
sum=10  
sum=15
```

Loops

pass statement

- ➔ For loops cannot be empty, but if you for some reason have a for loop with no content, put in the pass statement to avoid getting an error.

Example:

```
for l in 'Python':  
    pass
```

Output:

Loops

Break and continue statement

- ➔ With the **break** statement we can stop the loop before it has looped through all the items.
- ➔ With the **continue** statement we can stop the current iteration of the loop, and continue with the next.

Example:

```
for l in 'Python':  
    if l == 'h':  
        break  
    print("Current Letter:", l)
```

Output:

```
Current Letter:P  
Current Letter:y  
Current letter:t
```

Example:

```
for l in 'Python':  
    if l=='h':  
        continue  
    print("Current Letter:", l)
```

Output:

```
Current Letter: P  
Current Letter: y  
Current Letter: t  
Current Letter: o  
Current Letter: n
```

Loops

Else in for loop

- ➔ The else keyword in a for loop specifies a block of code to be executed when the loop is finished.

Example:

```
for x in range(6):  
    print (x, end=" ")  
else:  
    print("Finally finished!")
```

Output:

```
0 1 2 3 4 5 Finally finished!
```

Loops

Nested for loop

- ➔ A nested loop is a loop inside a loop
- ➔ The “inner loop” will be executed one time for each iteration of the “outer loop”.

Example:

```
i=["red", "blue", "brown"]  
j=["apple", "banana", "grape"]  
for b in i:  
    for a in j:  
        print(b,a)
```

Output:

```
red apple  
red banana  
red grape  
blue apple  
blue banana  
blue grape  
brown apple  
brown banana  
brown grape
```

Exercises

- <https://pynative.com/python-operators-and-expression-quiz/>
- <https://pynative.com/python-variables-and-data-types-quiz/>
- <https://pynative.com/basic-python-quiz-for-beginners/>
- <https://pynative.com/python/quizzes/>
- <https://pynative.com/python-basic-exercise-for-beginners/>

Thank You